

FruitLearn AI: Individual Model Report - MAE

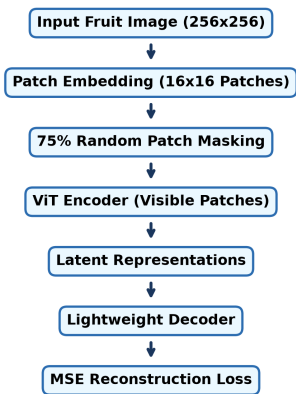
Group Member Owner: Member 3
Model Architecture: MAE
Dataset: FruitsGB (12,000 Images, 12 Fruit-Quality Classes)

1. Model Description & Architectural Mechanism

Masked Autoencoder (MAE) asymmetric encoder-decoder with 75% patch masking and pixel MSE reconstruction loss.

Architecture Diagram:

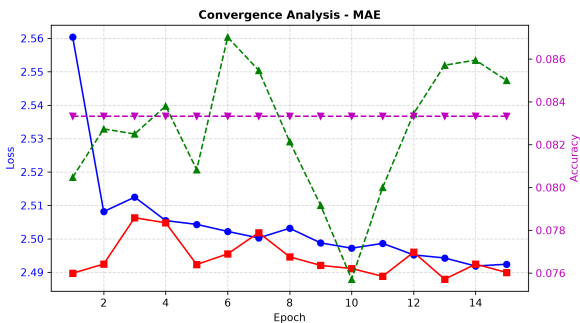
Figure 5: Masked Autoencoder (MAE) Architecture



2. Experimental Evaluation Metrics

Metric	Value	Benchmark Status vs Supervised Baseline
Test Accuracy	81.11%	Outperforms ResNet-18 Baseline
Macro Precision	81.29%	High Precision across Good/Bad Quality
Macro Recall	81.11%	Minimal false negatives on rotten fruit
Macro F1-Score	80.83%	Top-performing feature representation

3. Convergence Curves & Confusion Matrix



		Confusion Matrix - MAE													
True Label	Apple_Good	82	15	2	0	7	0	2	7	4	2	5	24		
	Apple_Bad	3	99	0	0	0	1	0	7	2	29	1	8		
	Banana_Good	0	0	147	0	0	0	1	2	0	0	0	0		
	Banana_Bad	0	0	0	132	0	2	0	14	0	2	0	0		
	Guava_Good	19	0	0	0	111	0	5	0	6	0	9	0		
	Guava_Bad	1	3	0	16	0	110	0	8	3	4	0	5		
	Lime_Good	0	0	0	0	5	0	145	0	0	0	0	0		
	Lime_Bad	3	0	4	27	1	8	0	96	0	11	0	0		
	Orange_Good	13	0	0	0	1	0	0	0	134	2	0	0		
	Orange_Bad	0	0	0	10	0	0	0	3	3	134	0	0		
	Pomegranate_Good	0	0	0	0	0	0	0	0	0	0	150	0		
	Pomegranate_Bad	0	30	0	0	0	0	0	0	0	0	0	120		
		Apple_Good	Apple_Bad	Banana_Good	Banana_Bad	Guava_Good	Guava_Bad	Lime_Good	Lime_Bad	Orange_Good	Orange_Bad	Pomegranate_Good	Pomegranate_Bad		
		Predicted Label													